Caitlin Roake, Joely Marsyla, Natalie Zorn

Milestone 1 and Peer Feedback

Project description

We propose using a database of the narrative ANAC reports to investigate trends in mountaineering accidents over 30 years of data. Our primary objectives are:

- To describe trends in proximal accident cause over time (AAC categories include fall on rock, rappel error, avalanche, exposure, rock fall)
- To map the geographic distributions of mountaineering accidents
- To compare the number of accidents per discipline (rock, ice, alpine, mountaineering) over time against timelines of introductions of novel safety gear (for example camming devices for rock protection, dynamic ropes, helmets)
- To query whether accident severity can be predicted by discipline, prior experience of the participants, time of year, proximal cause of the accident, and equipment used, or some combination of these factors.

Data description

Our dataset comes from a publicly available GitHub repository: https://github.com/ecaroom/climbing-accidents

The repository is openly accessible and licensed for public use. The data is stored in an excel file containing structured data that documents climbing accident incidents. Each row corresponds to a single recorded incident, and each column represents an attribute that describes the accident.

The dataset contains 2,770 rows, each representing a climbing accident. The first few columns in the dataset describe the accident, year, additional descriptive information, and tags that were applied to the accident. The additional attributes describing each accident includes:

- Date of incident
- Location
- Age Range
- Ability level
- Injury type
- Type of climbing
- Cause of the accident
- Weather/site conditions during the accident
- Equipment involved

There are additional tabs included in the dataset including the total incident count, roped count by style, and TR-sport-trad ex-ALP. Because the dataset includes location information, this allows for spatial analysis and different visualization techniques described below. The data includes dates and categorical variables that can allow us to compare and classify different features for our analysis.

Ethical data concerns

This project directly addresses individuals partaking in mountaineering activities, including varied disciplines of climbing, skiing, and traversing. This dataset contains reported incidents in mountaineering, where individuals may be identified based on unique circumstances of the occurrence. However, these incidents were self-reported, so their consent was most likely required for publication. A harmful outcome of this project could be public perception on outdoor activities, where some might be discouraged in participating.

Alpine guides on gear, safety, and recommended practices might also be influenced by our project, where trends in incidents may reveal holes in the widely accepted principles. Popular organizations that publish include the American Alpine Club, International Climbing and Mountaineering Federation, and International Federation of Mountain Guides Association. These are globally renowned organizations that prioritize safe practices that reduce injuries in alpine settings, advising gear, route selection, and activity choice.

Companies that manufacture gear uphold to the highest standards, prioritizing safety and minimizing liability surrounding the brand and produced gear. These companies could be considered stakeholders, as improper user instruction or faulty mechanisms might influence the overlap between gear safety and operations. However, this project does not intend to influence perception on individual brands or companies, but if a product is mentioned in the report, it should be considered if it is listed as the reason for injury.

An additional stakeholder is medical professionals and search/rescue teams. This project could advise efficient responses, adequate training and funding, and the availability of resources. These are relevant to reactions but limited in preventing occurrences.

Methods

Temporal Extraction (Month)

We extracted the month of each incident from narrative text using regular expressions. The

parser searched for full, capitalized month names (e.g., “January”–“December”) and returned the first occurrence. Extracted months were then converted to numerical format (1–12). Entries without a recognizable month were recorded as missing.

Geographic Extraction (State/Province and Country)

Location information was parsed from the text fields by matching known place names against a reference list of U.S. states, Mexican states, Canadian provinces, and countries. When multiple locations were present, priority was given to the most specific identifiable administrative unit (state over country). Extracted locations were standardized to consistent naming conventions for downstream analysis.

Geospatial Coordinates (Mountain Project Integration)

We supplemented location data using a Mountain Project dataset containing route names and associated latitude/longitude coordinates. Incident records were matched to this dataset via route or area names using string matching. When a match was identified, the corresponding coordinates were assigned to the incident. Unmatched records retained missing coordinate values.

Injury Severity Classification

Injury severity was assigned using a rule-based natural language processing approach. Narrative text was scanned for predefined keyword patterns corresponding to four categories: *fatal*, *serious*, *moderate*, and *minor*. Patterns included terms such as “fatal,” “unconscious,” “fracture,” and “minor injuries,” with additional phrases tailored to the dataset (e.g., “body recovered,” “evacuated,” “hypothermia”). Each record was assigned to a severity label based on the highest-priority category detected. For transparency and iterative refinement, matched terms were also recorded for each entry.

Climber Experience Classification

We will infer climber experience level (novice, intermediate, expert) from narrative reports using a rule-based natural language processing approach. Text will be scanned for indicators such as years of experience, prior climbs, technical grades, and self-described ability (e.g., “first lead,” “experienced guide,” “advanced climber”). These indicators will be mapped to discrete categories using a predefined hierarchy. In ambiguous cases, multiple signals will be aggregated and the highest-confidence category assigned. Extracted labels will be manually reviewed on a subset of records to refine classification rules.

Climbing Discipline Classification

Mountaineering and climbing are broad classifications of largely interchangeable practices. Mountaineering generally refers to the sport of climbing mountains combining techniques from rock climbing, ice climbing, glissading, traversing, route selection, and equipment use. Rock climbing is a broad classification of many disciplines with protection and risk defining each. Sport climbing refers to roped climbing where permanent protection drilled into the rock provides the least risk associated with pitch climbing. Traditional or trad climbing refers to roped climbing with temporary gear placed in the rock by the climber, offering some protection but a greater risk than sport. Bouldering refers to unroped climbing on large boulders, typically involving crash pads beneath the climber to prevent hard falls to the ground. Ice climbing is a winter activity where climbers solo/climb unroped, or drill temporary protection into the ice for roped climbing. Classification of these disciplines proved difficult, as the verbiage is extremely similar, resulting in overlap in some of the tags. A rule-based language processing supported the categorization of incidents to “alpine/mountaineering”, “rock” and “ice” climbing. General climbing vernacular provided unclear tags for each incident, so some incidents were individually analyzed for specific keywords. An ordering system was included to prioritize “ice”, then “rock”, then “alpine” to hinder the overflow into the alpine category.

Rope Classification

While climbing with a rope is almost always encouraged, some climbs/climbers opt not to utilize one for protection. Especially in rock climbing, the use of a rope can greatly decrease the risk of fatality of falls. Similar to the tagging mentioned above, a rule-based language classification approach allowed for the sorting of each incident to “roped” or “unroped.” Some incidents had no mention of ropes/other protection, and they reside in an “unknown” category.

Helmet Classification

Helmets provide the most comprehensive cranial protection for climbers, alpinists, and skiers alike. Despite this knowledge, an overwhelming majority of incidents in the dataset describe climbers not wearing one. Many incidents have no mention of helmets, so there is a disproportionate number of unknown incidents, especially those where minor injuries not involving the head are reported.

Feature Encoding for Machine Learning

Categorical variables (e.g., month, location, route type, severity tags, and inferred experience level) will be transformed into numerical representations suitable for machine learning. Nominal variables will be encoded using one-hot encoding, while ordinal

variables (e.g., experience level) will be encoded using ordered integer mappings. Missing values will be handled through imputation or inclusion of explicit "unknown" categories. All features will be standardized or normalized as appropriate for the selected modeling approach.

Predictive Modeling of Accident Severity

We will train supervised machine learning models to predict accident severity (deadly, serious, moderate, minor) using the engineered feature set. Candidate models will include logistic regression, random forests, and gradient boosting classifiers. The dataset will be split into training and test sets, with cross-validation used to tune hyperparameters and reduce overfitting. Model performance will be evaluated using metrics appropriate for multiclass classification, including accuracy, precision, recall, F1 score, and confusion matrices. Feature importance analyses will be conducted to identify the most influential predictors of severity.

<u>Preliminary results</u>

As discussed in the summary below, all code files are included in the GitHub repository.

Spatial Analysis: The map of climbing accidents across North America shows clear clustering rather than an even spread. Most accidents are concentrated in the western United States, especially in states like California, Colorado, Utah, and Washington. This lines up with where major climbing areas are located, which suggests that accident frequency is closely tied to where people are actually climbing. This is shown in the discipline-based scatter plot shown in Figure 1 below. This map is also interactive on Python. When moving the mouse over the points, the user can view the exact accidents, its location, and year.

Rock climbing accounts for the majority of incidents and appears across almost all regions in the dataset. This isn't surprising given how popular and widespread rock climbing is compared to other disciplines. The densest clusters of rock-related accidents are in popular climbing areas in the western parts of the US.

Alpine and mountaineering accidents are more limited. These tend to show up in higher elevation regions such as the Rockies, the Pacific Northwest, and Alaska. This makes sense since these activities depend on specific terrain and conditions.

Ice climbing accidents are the least common and are mostly concentrated in northern and colder regions. Their distribution is much more localized, which likely reflects both the seasonal nature of ice climbing and the smaller number of participants.

Overall, the spatial patterns suggest that accidents are primarily driven by where climbing activity is concentrated, rather than being randomly distributed.

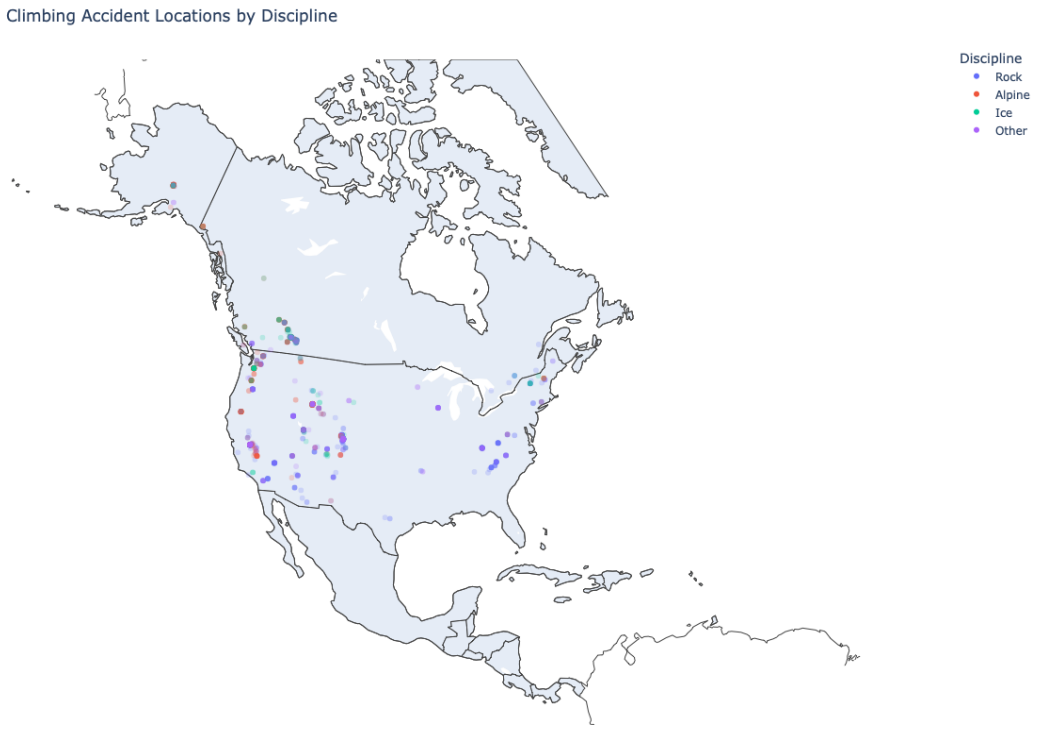

Figure 1: Climbing Accident Locations by Discipline (Rock, Alpine, Ice, & Other)

Looking at accident severity adds another layer to the spatial patterns. Incidents of all severity levels appear in similar regions. These points can be observed in Figure 2 below.

More severe accidents, classified as “Deadly” or “Serious”, appear more often in mountainous and remote areas, especially in the western US and parts of Canada. These regions are associated with alpine terrain and more technical routes, which could help explain why accidents there are more severe.

Less severe incidents, such as “Minor” and “No/Little,” are more spread out and appear in both major climbing areas and more accessible locations. This could be because these types of accidents are more common overall, or because they are more likely to be reported in a wider range of settings.

There isn’t a single location that stands out as uniquely dangerous in terms of severity. Instead, severity seems to follow the same general spatial patterns as overall accident frequency, with some increase in more remote or technically challenging areas.

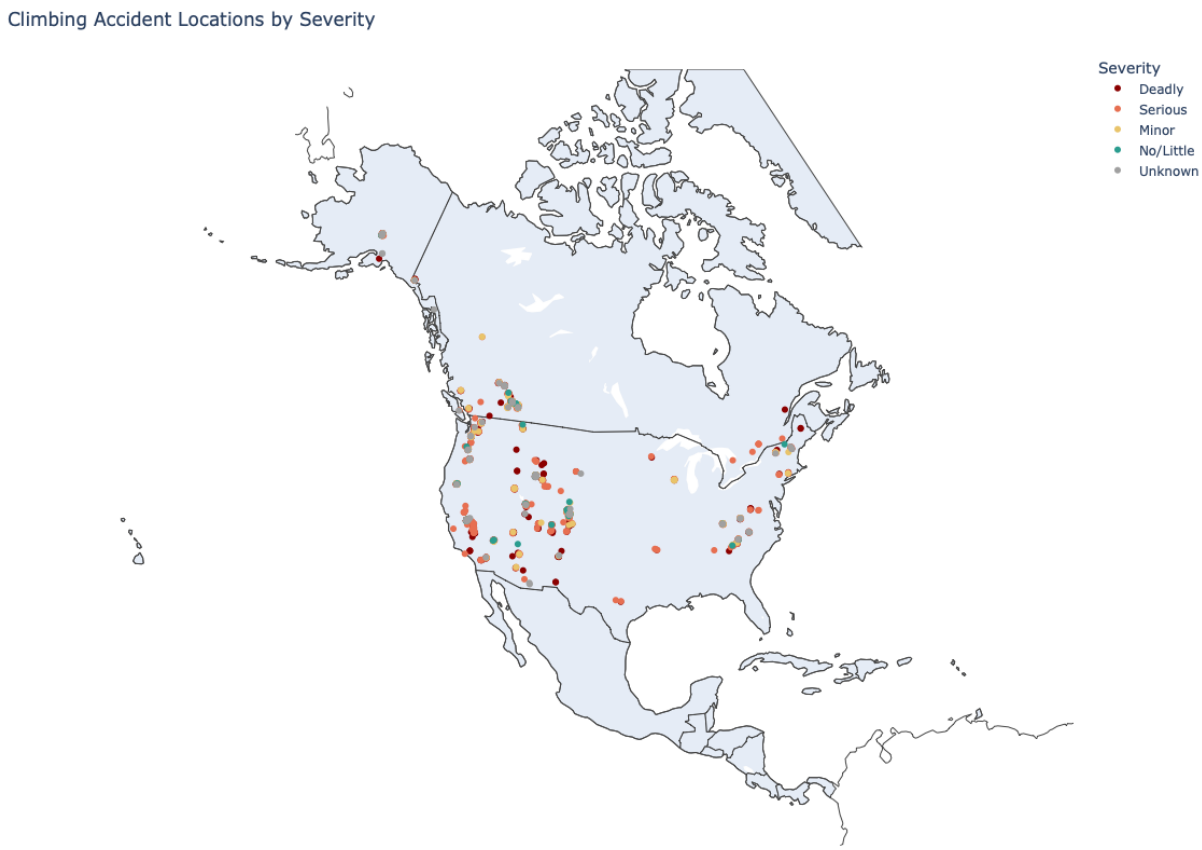

Figure 2: Climbing Accident Locations by Severity (Deadly, Serious, Minor, No/Little, Unknown)

Temporal Analysis: This next section looks at temporal trends to better understand when accidents are most likely to occur.

The yearly accident counts stay high across the entire time period, but they do not increase or decrease in a steady way. The highest incident count occurs around mid to late 2010s, before dropping steeply around 2020. The decrease at the end likely reflects incomplete reporting or changes in the data collection, rather than a true decline in incidents. The histogram in Figure 3 above shows that most yearly incident totals fall within a narrow range, with some years standing out as unusually low or high. The data shows fluctuations from year to year, with a dip in the mid to late 2000s. The data rises again in the 2010s. These variations are observed in Figure 3 and Figure 4 below. The overall pattern does show a small upward trend over time as shown in Figure 4. The fitted linear trend line in the plot shows a small positive slope that was calculated to be 0.0594.

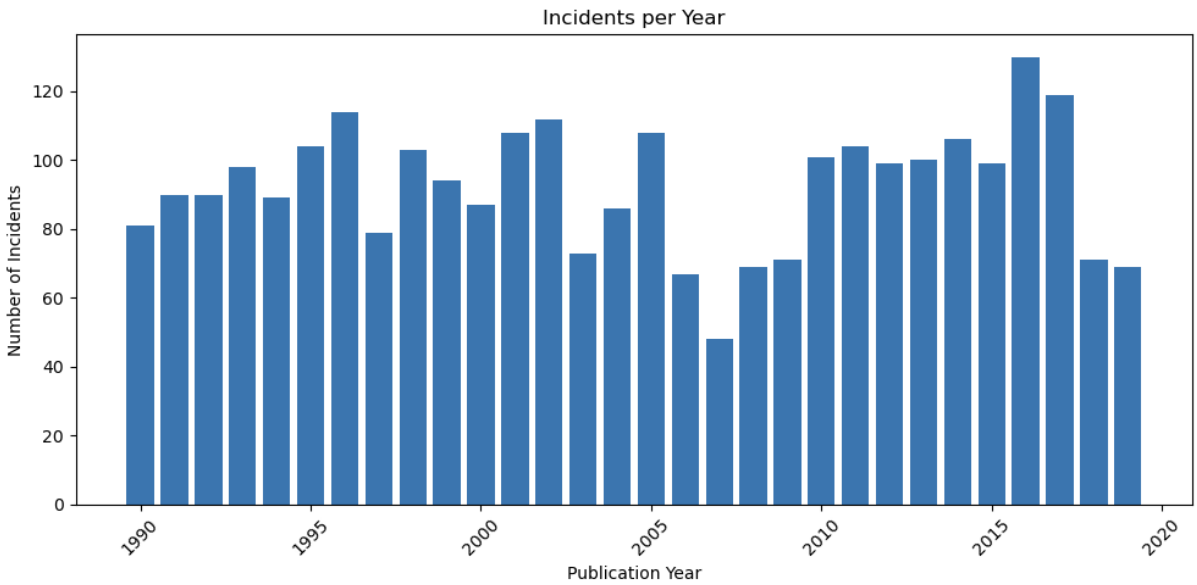

Figure 3: Incidents per Year Bar Chart

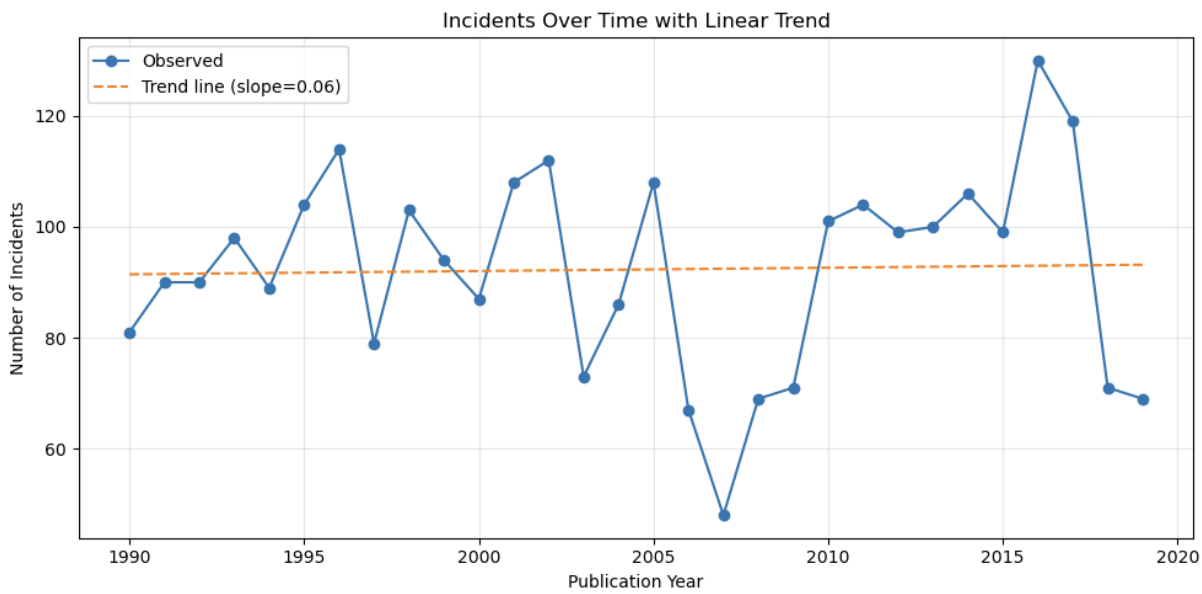

Figure 4: Line Plot Showing Trend in Incidents Over Time

Figure 5 below shows the number of incidents broken down by discipline. Rock climbing has the highest number of incidents in most years. This makes sense because rock-climbing incidents are most prevalent when looking at the dataset. Alpine incidents are also present; they show some variability over time but remains relatively constant with time. Ice and Other incidents are much lower and don't fluctuate as much. Overall, the discipline trends show that the total incident counts are largely driven by rock climbing and alpine.

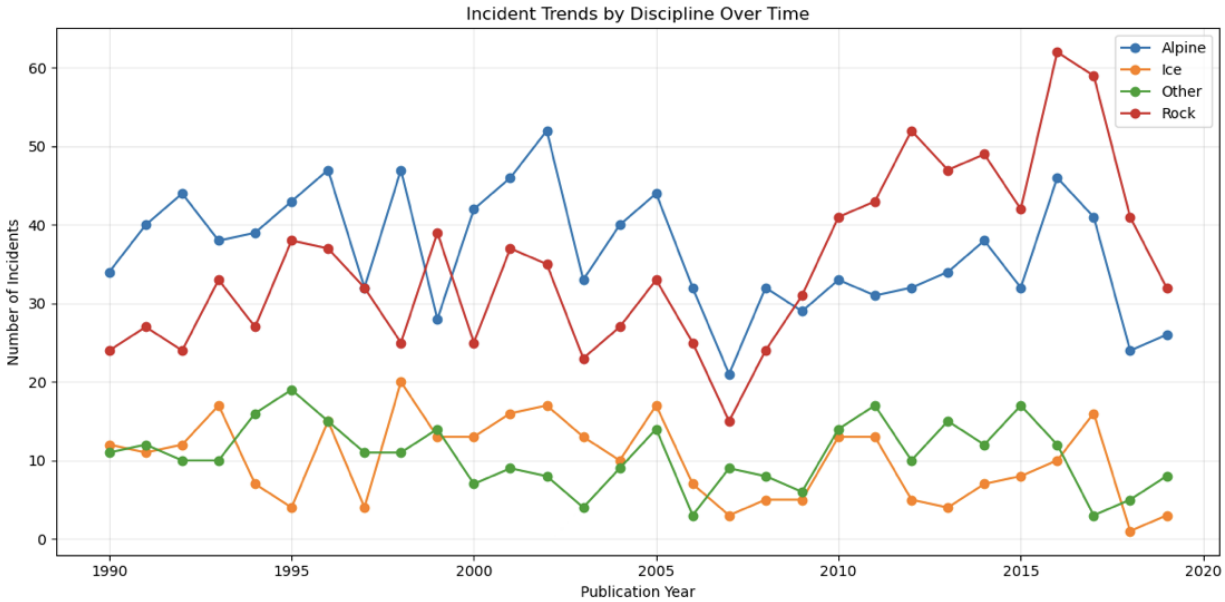

Figure 5: Incident Trends by Discipline Over Time

Figure 6 below shows a similar line plot but compares severity over time. The plot shows that serious and fatal accidents make up most of the reported incidents. These categories remain consistently higher than the minor, moderate, and unknown categories. Moderate and minor incidents occur less frequently, and the "unknown" category remains low across nearly all years. This may reflect reporting bias, where more severe incidents are more likely to be documented. This difference is visible in the figure below.

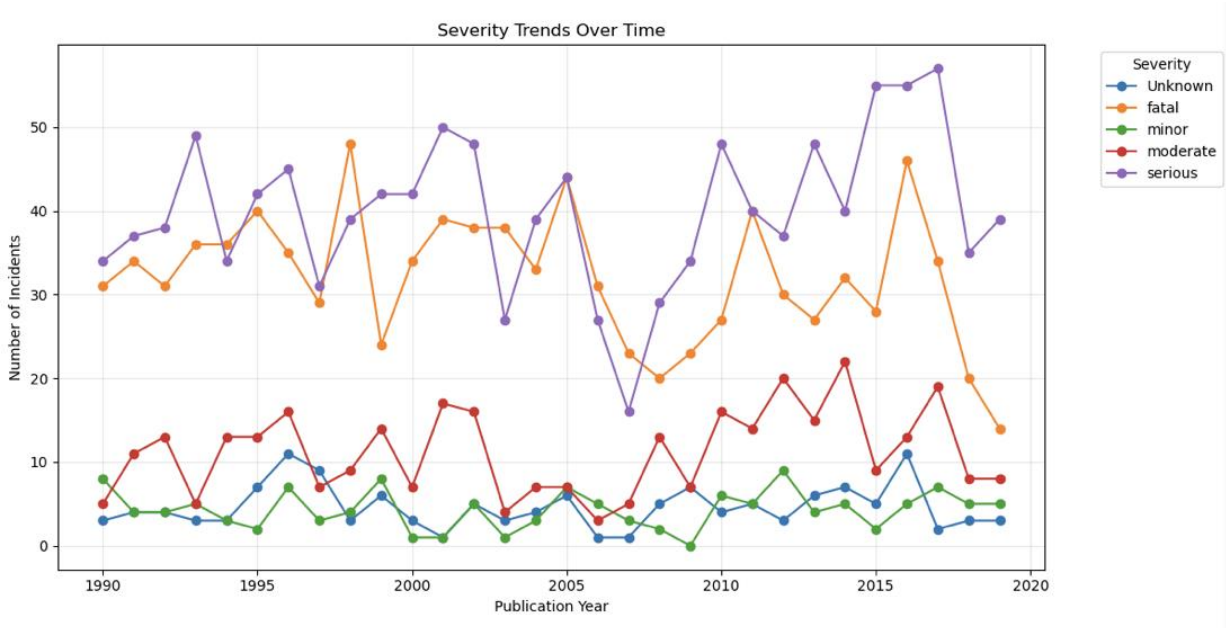

Figure 6: Severity Trends Over Time

Together with the spatial and temporal analyses, climbing accidents are observed to be concentrated both geographically and across consistent long-term trends. These prelimnary results provide a good foundation for deeper analyses like factors that influence accident frequency and severity.

Peer Feedback

Our team present: Joely Marsyla, Caitlin Roake, Natalie Zorn: An Analysis of 30 Years of Self-reported Mountaineering Accidents

Othe team present: Cooper Kerr, Minh Le: Formula 1 Predictive Analysis

Feedback we received: Our feedback group suggested that we use other data sources to add more granular location data to our analysis. In response to this feedback, we used a database from the website mountain project to extract GPS coordinates based on popular climbing areas. They also suggested that we focus on one main outcome. We decided as a group to work towards a model to determine which predictors (time of year, climbing discipline, experience level) could be used to predict injury severity. The quality of the feedback we received was excellent.

Feedback we gave:

- Consider finding ways to clean/merge other datasets, so your project incorporates more data cleaning and processing.
- Add other tags to your current dataset to perform multiple analyses.

A List of completed milestones

- Exploratory Analysis – completed
- Temporal Analysis – completed

A List of Methods milestones

- Temporal Extraction (Month) - completed
- Geographic Extraction (State/Province and Country) - completed
- Geospatial Coordinates (Mountain Project Integration) - completed
- Injury Severity Classification – completed
- Climbing Classification – completed
- Helmet Classification – completed

A List of milestones to complete

- Analyze correlations between equipment advancement and injury patterns/frequency. This analysis will allow us to see any correlations between safety advances and reduction in accidents.
- Rope Classification – mostly completed
- Cause of incident Classification – mostly completed

Summary

A link to our GitHub repository has been provided, with multiple python files accomplishing data classifications and analysis. The file “"ANAC_mp_loc.csv" contains the comprehensive dataset sourced from the American Alpine Club. Multiple files include tagging and classifications for the methods described above, “”ANAC_severity.ipynb”, “ANAC_loc.csv”, and “ANAC_climb.py”. The file “ANAC_date.ipynb” shows temporal analysis and “Spatial Analysis - LATEST Update.ipynb 2.zip” contains spatial analysis.

This report highlights the methodology of data acquisition and cleaning, preliminary results, and future milestones. Our methodology of data acquisition and cleansing describes the extensive nature of classification for future analysis. The dataset is extensive and reflects the broad sourcing of each incident. Each incident mainly highlights injuries and cause of occurence, but not necessarily the activity/safety precautions of individuals. Predictive modelling is an interesting caveat of this project, where medical teams and first responders might benefit patient care and extraction of injured persons.

Preliminary results with spatial and temporal data visualizations have been provided, showing the distribution of incidents and prevalence among established sporting areas. Future analysis will build upon this data to find trends in climbing patterns, standardized practices, and medical responses.